ACADEMIC YEAR: 2025-26

Course: Skill base Lab Course: Cloud Computing

Course code: CSL605

Year/Sem: TE/VI

Experiment No.: 03
Aim: To study and Implement Platform as a Service using AWS Elastic Beanstalk/ Microsoft Azure App Service
Name: Shravani Sandeep Raut
Roll Number: 40
Date of Performance: 07/01/2026
Date of Submission: 12/01/2026

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	**20**	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	2
Understanding	4-5	2-3	2
Journal work and timely submission.	8-10	5-8	4

Checked by

Name of Faculty : Dr. Tatwadarshi Nagarhalli

Signature :

Date :

Experiment No. 3

Aim: To study and Implement Platform as a Service using AWS Elastic Beanstalk/ Microsoft Azure App Service

Objective: Understand the concept of PaaS and implement using Own Cloud which gives universal access to files through a web interface.

Theory:

- PaaS is a platform for programming developers and brings benefits - SaaS is used for but from the software development point.
- It is the computer platform that provides the facility to use web applications quickly. With ease, without buying & maintaining web-development, PaaS has a similarity with that SaaS except that SaaS delivers software over the web.
- In contrast, PaaS provides a platform for the creating of software delivered over the web.
- PaaS has a feature of a point-and-click tool that allows non-programmers to develop web applications.
- App-Engine of Google & Force.com, Windows Azure, AppFog, Openshift, and VMware Cloud Foundry are PaaS examples.
- <u>Advantages of PaaS:</u>
    - Scalability: of users ranges from hundreds to thousands.
    - Prebuilt Business Plan: PaaS vendors provide pre-defined business functionality for users to directly start the project.
    - Low Cost: Development via PaaS requires a computer & a good internet connection and less investment in hardware & software.
    - Instant Community: PaaS providers facilitates user providing online communities where a developer can get new ideas & share their experience & advice.
    - Simple & easy to use
- <u>Disadvantages of PaaS are as follows:</u>
    - Vendor Migration: Migration from one PaaS vendors' application to another PaaS vendor will create some problem.
    - Data-Privacy: The privacy of data can get hamper if it is not held within the company's boundary or organization.
    - Mix-up Complexity: Some of the applications developed may be local while others are from the cloud, which may increase the complexity.

Steps:

Step1 : Login to AWS console and go to Elastic Beanstalk

Step 2: Click on Create Application

Step 3: Write Application information : Name, Tag,Platform

etc. Step 3: Write Application information : Name,

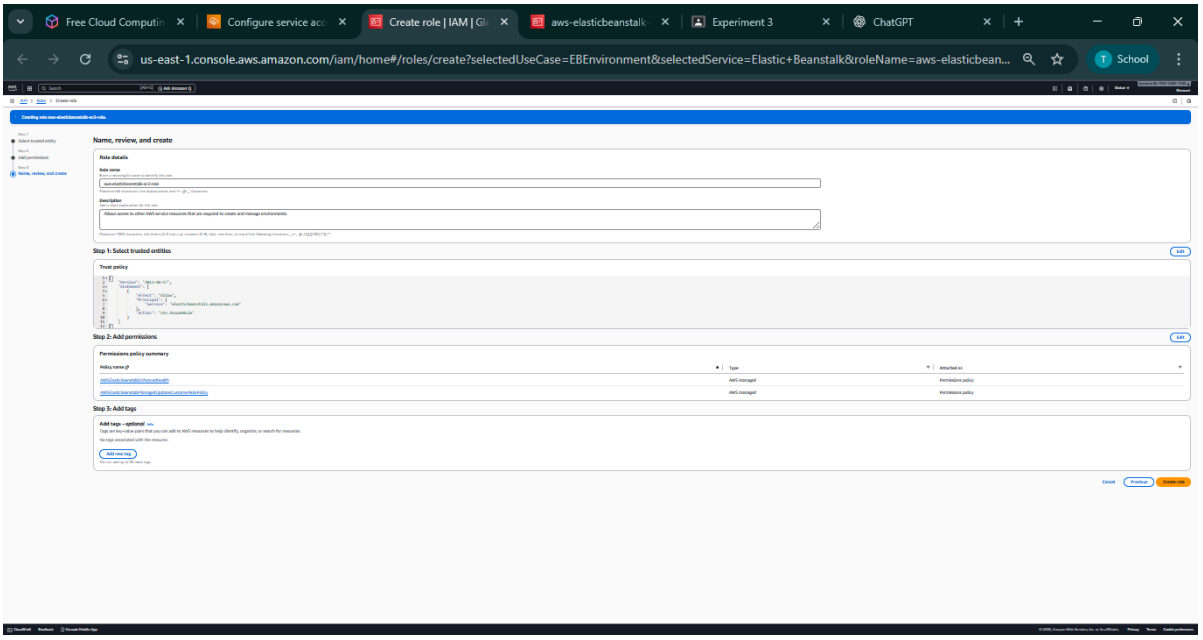
Tag,Platform etc. This will take a few minutes.

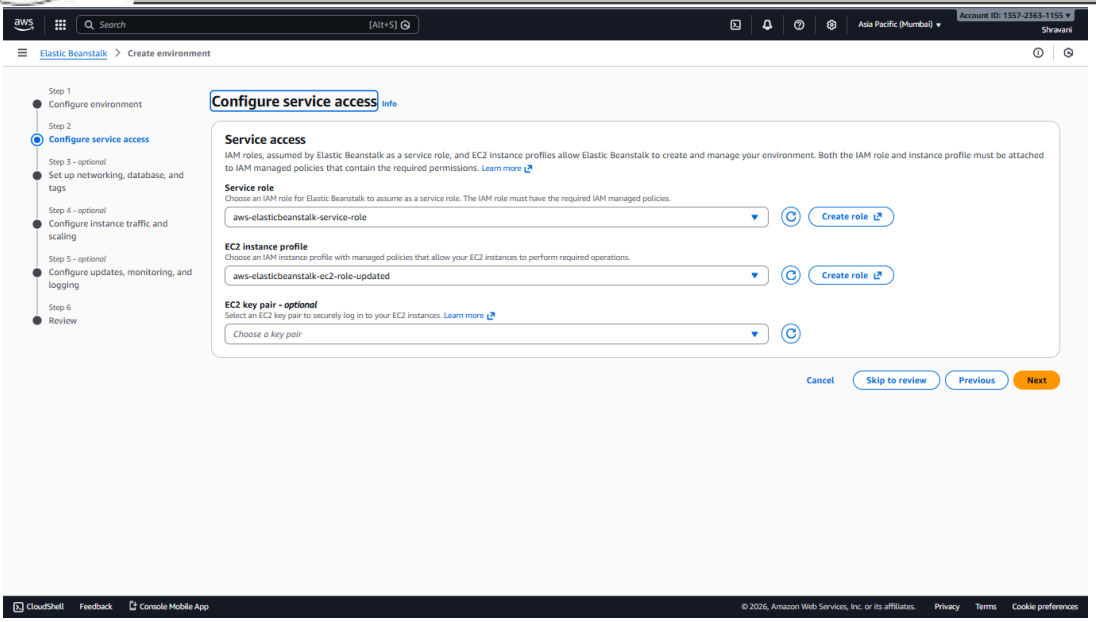
Step 5: Click on Environments -> Check the health of Environment wait till it becomes

'OK Step 6: Click the URL

To Delete the application and Environment (Select it and in Action -Delete/Terminate : give conformation

Output/Observation:




Elastic Beanstalk > Create environment
Step 1
Configure environment
Step 2
Configure service access
Step 3 - optional
Set up networking, database, and tags
Step 4 - optional
Configure instance traffic and scaling
Step 5 - optional
Configure updates, monitoring, and logging
Step 6
Review
Configure service access
Info
Service access
IAM roles, assumed by Elastic Beanstalk as a service role, and EC2 instance profiles allow Elastic Beanstalk to create and manage your environment. Both the IAM role and instance profile must be attached to IAM managed policies that contain the required permissions.
Learn more
Service role
Choose an IAM role for Elastic Beanstalk to assume as a service role. The IAM role must have the required IAM managed policies.
aws-elasticbeanstalk-service-role
Create role
EC2 instance profile
Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.
aws-elasticbeanstalk-ec2-role-updated
Create role
EC2 key pair - optional
Select an EC2 key pair to securely log in to your EC2 instances.
Learn more
Choose a key pair
Cancel
Skip to review
Previous
Next
CloudShell
Feedback
Console Mobile App
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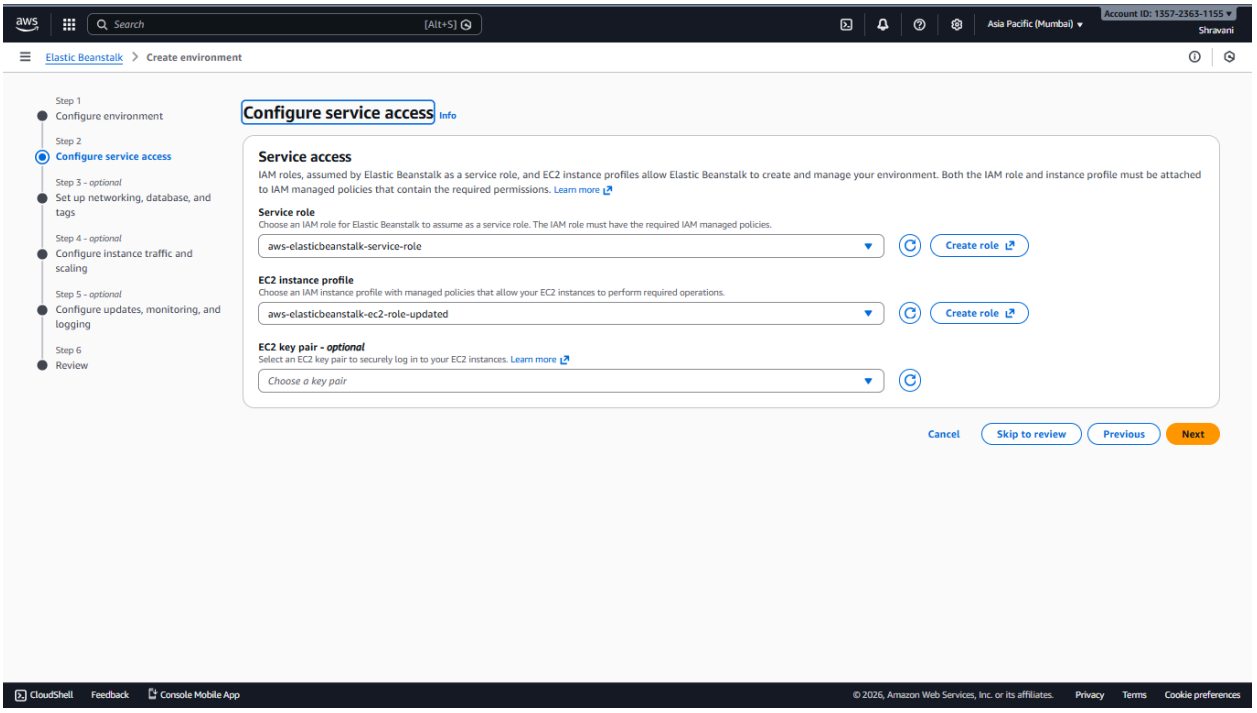
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EC2 instance profile
Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.
aws-elasticbeanstalk-ec2-role-updated
Create role
EC2 key pair - optional
Select an EC2 key pair to securely log in to your EC2 instances.
Learn more
Choose a key pair
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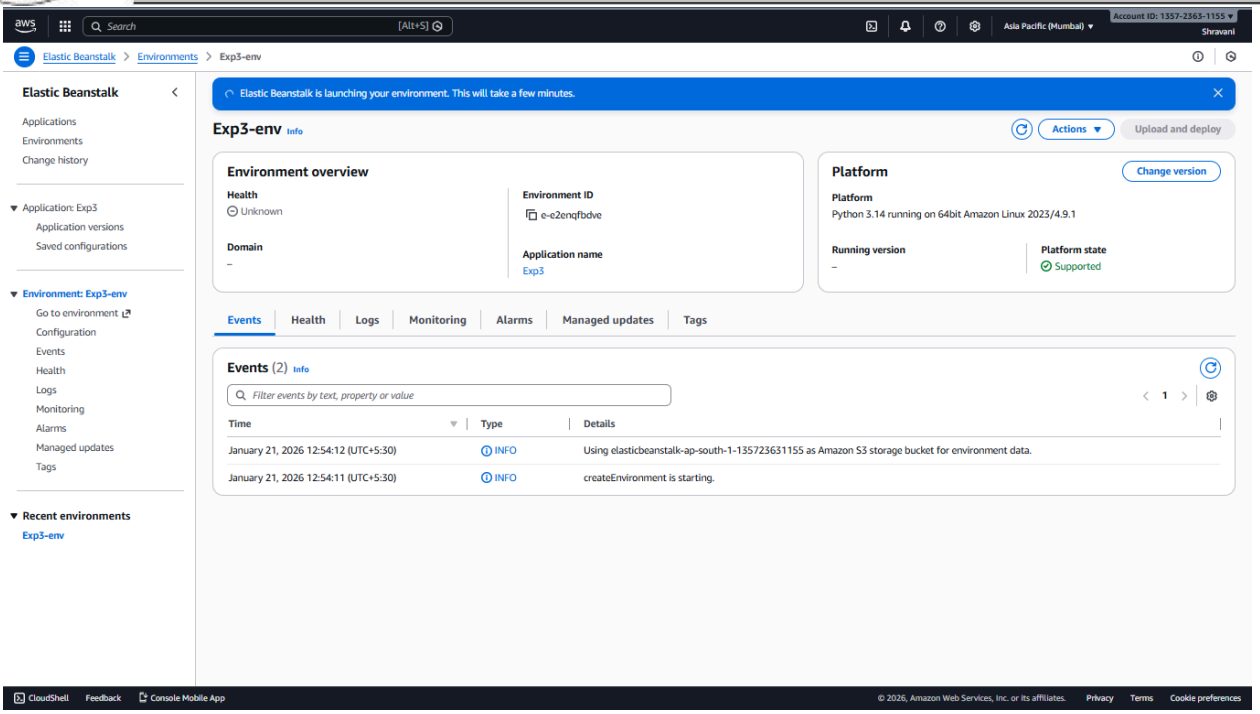
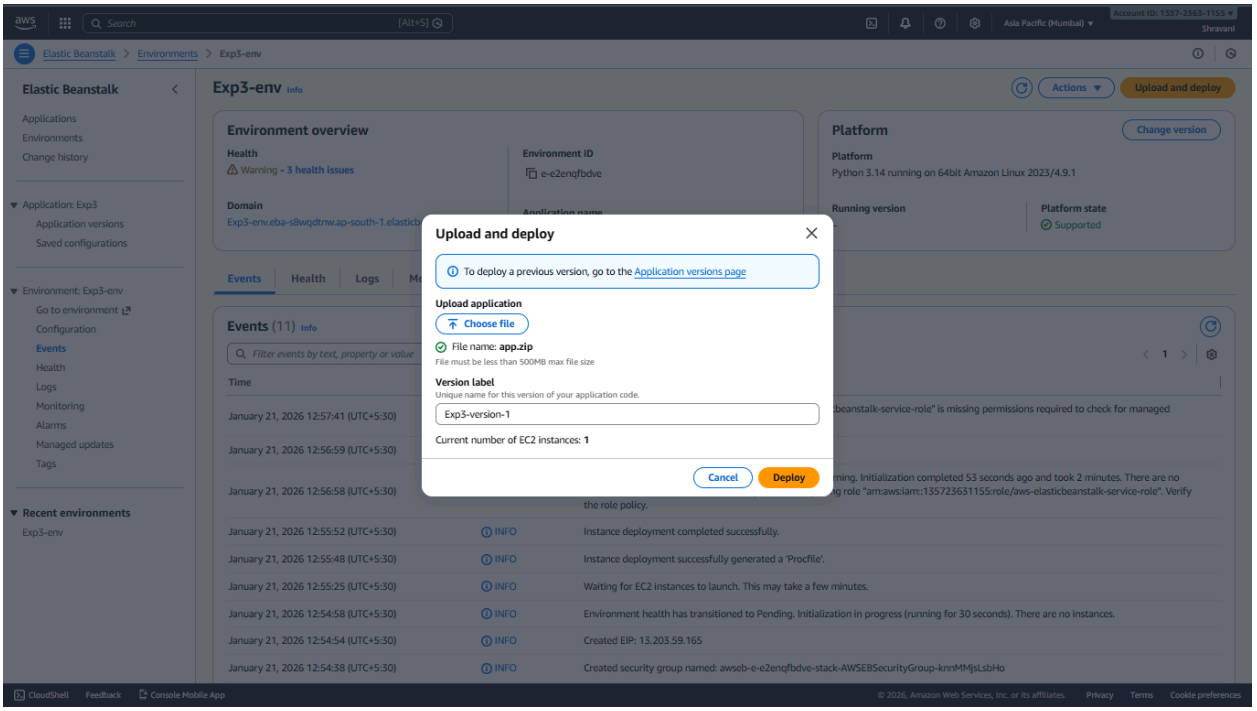
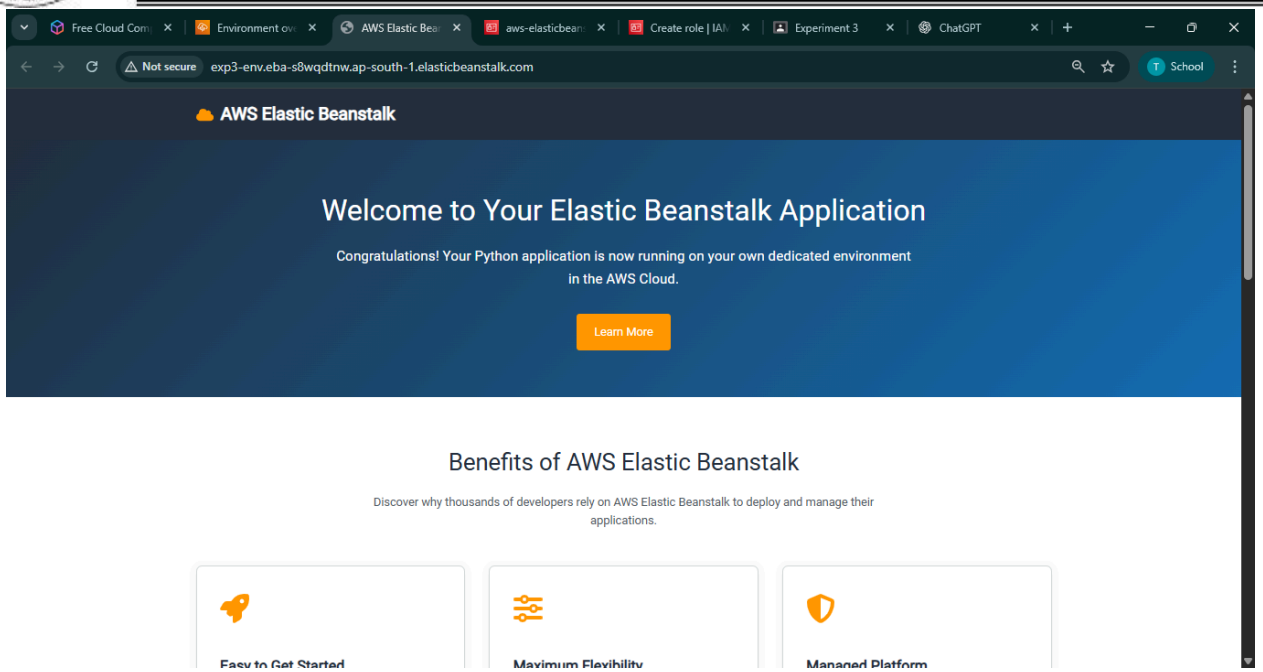

Elastic Beanstalk > Environments > Exp3-env
Elastic Beanstalk is launching your environment. This will take a few minutes.
Exp3-env
Actions
Upload and deploy
Environment overview
Health
Unknown
Environment ID
e-e2enqfbdve
Domain
–
Application name
Exp3
Platform
Change version
Platform
Python 3.14 running on 64bit Amazon Linux 2023/4.9.1
Running version
–
Platform state
Supported
Events
Health
Logs
Monitoring
Alarms
Managed updates
Tags
Events (2)
Time
Type
Details
January 21, 2026 12:54:12 (UTC+5:30)
INFO
Using elasticbeanstalk-ap-south-1-135723631155 as Amazon S3 storage bucket for environment data.
January 21, 2026 12:54:11 (UTC+5:30)
INFO
createEnvironment is starting.


Elastic Beanstalk > Environments > Exp3-env
Exp3-env
Actions
Upload and deploy
Environment overview
Health
Warning - 3 health issues
Environment ID
e-e2enqfbdve
Domain
Platform
Change version
Python 3.14 running on 64bit Amazon Linux 2023/4.9.1
Running version
Platform state
Supported
Upload and deploy
To deploy a previous version, go to the Application versions page
Upload application
Choose file
File name: app.zip
File must be less than 500MB max file size
Version label
Unique name for this version of your application code.
Exp3-version-1
Current number of EC2 instances: 1
Cancel
Deploy
Events (11)
January 21, 2026 12:55:52 (UTC+5:30)
INFO
Instance deployment completed successfully.
January 21, 2026 12:55:48 (UTC+5:30)
INFO
Instance deployment successfully generated a 'Procfile'.
January 21, 2026 12:55:25 (UTC+5:30)
INFO
Waiting for EC2 instances to launch. This may take a few minutes.
January 21, 2026 12:54:58 (UTC+5:30)
INFO
Environment health has transitioned to Pending. Initialization in progress (running for 30 seconds). There are no instances.
January 21, 2026 12:54:54 (UTC+5:30)
INFO
Created EIP: 13.203.59.165
January 21, 2026 12:54:38 (UTC+5:30)
INFO
Created security group named: awseb-e-e2enqfbdve-stack-AWSEBSecurityGroup-knnMMjsLsbHo



Conclusion:

PaaS does not require traditional installation. The cloud provider manages the entire platform setup, allowing developers to focus only on application development. This results in faster deployment, reduced maintenance, and improved efficiency